

Unit 7 Test Study Guide

(Polynomials & Factoring)

Name: _____

Date: _____ Per: _____

Topic 1: Writing Polynomials in Standard Form

Directions: Write each polynomial in standard form.

1. $a^2 + 3a^4 - 7a + 2a^3 - 4$

$$3a^4 + 2a^3 + a^2 - 7a - 4$$

2. $8y^3 - 3xy^2 - x^2y + 2x^3$

$$2x^3 - x^2y - 3xy^2 + 8y^3$$

Topic 2: Classifying Polynomials

Directions: Classify each polynomial by degree and number of terms.

3. $3x + 12$

linear, binomial

4. 24

constant, monomial

5. $-7x^2 + 4x + 1$

quadratic, trinomial

6. $3x^4 - x^3 + 5x^2 + x - 7$

quartic, polynomial

Topic 3: Simplifying Polynomials

Directions: Simplify each polynomial. Write all answers in standard form.

7. $(2x^2 + 3x + 2) - (x^2 - 4x - 1)$

$$x^2 + 7x + 3$$

8. $2a^3(5a^2 - 7a) + (3a^4 - a^5)$

$$10a^5 - 14a^4 + 3a^4 - a^5$$

$$9a^5 - 11a^4$$

9. $5c^2d(9c^2d^2 - 4c^3d - 2)$

$$45c^4d^3 - 20c^5d^2 - 10c^2d$$

$$-20c^5d^2 + 45c^4d^3 - 10c^2d$$

10. $-4y(5y - 11) - 2y(3y + 10)$

$$-20y^2 + 44y - 6y^2 - 20y$$

$$-26y^2 + 24y$$

11. $(x - 1)(4x - 3)$

$$4x^2 - 3x - 4x + 3$$

$$4x^2 - 7x + 3$$

12. $(5y - 1)^2 \rightarrow (5y - 1)(5y - 1)$

$$25y^2 - 5y - 5y + 1$$

$$25y^2 - 10y + 1$$

13. $(2x + 5y)(x - 3y)$

$$2x^2 - 6xy + 5xy - 15y^2$$

$$2x^2 - xy - 15y^2$$

14. $(x + 4)(4x - 2) - 2x^2$

$$4x^2 - 2x + 16x - 8 - 2x^2$$

$$2x^2 + 14x - 8$$

15. $\frac{-8x^5 + 28x^4 - 4x^3}{4x^3}$

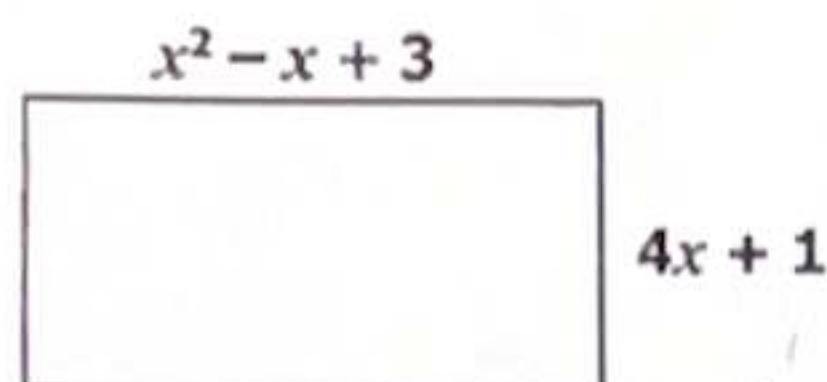
$$-2x^2 + 7x - 1$$

16. $\frac{27c^4d^7 + 15c^3d^5}{3c^2d^3}$

$$9c^2d^4 + 5cd^2$$

Topic 4: Geometric Applications

17. Write an expression to represent the area of the rectangle below in simplest form.

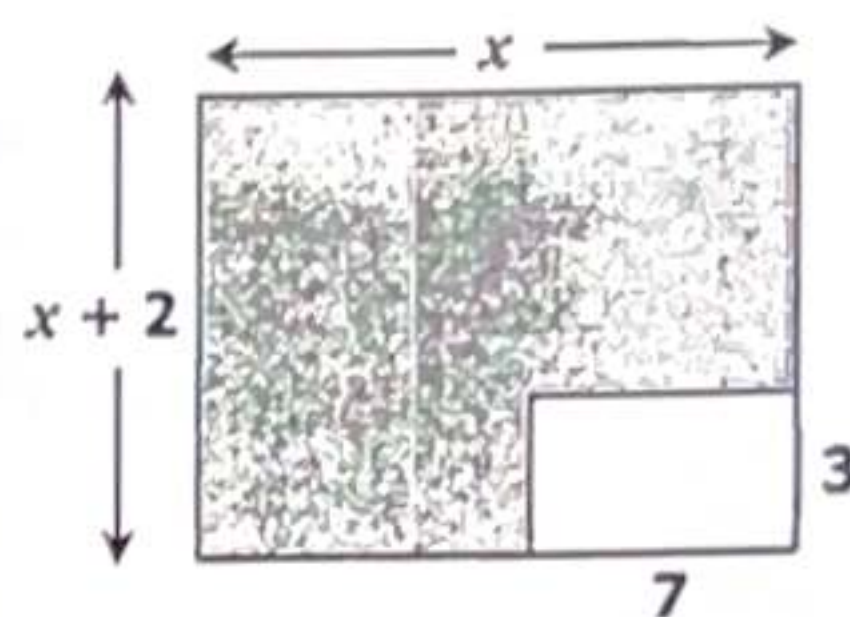


$$(x^2 - x + 3)(4x + 1)$$

$$4x^3 + x^2 - 4x^2 - x + 12x + 3$$

$$4x^3 - 3x^2 + 11x + 3$$

18. Write an expression to represent the area of the shaded region in simplest form.



$$x(x+2) - 7(3)$$

$$x^2 + 2x - 21$$

Topic 5: Factoring Polynomials

Directions: Factor each polynomial completely.

19. $54x^3y - 18x^2y^2 + 27xy^3$

$$9xy(6x^2 - 2xy + 3y)$$

20. $9x^2 - 49y^2$

$$(3x + 7y)(3x - 7y)$$

21. $64y^4 - 16y^6$

$$16y^4(4 - y^2)$$

$$16y^4(2 + y)(2 - y)$$

22. $n^3 - 5n^2 - 50n$

$$n(n^2 - 5n - 50)$$

$$n(n - 10)(n + 5)$$

23. $4x^3 + 12x^2 - 25x - 75$

$$4x^2(x + 3) - 25(x + 3)$$

$$(4x^2 - 25)(x + 3)$$

$$(2x + 5)(2x - 5)(x + 3)$$

24. $3k^2 - 57k + 180$

$$3(k^2 - 19k + 60)$$

$$3(k - 15)(k - 4)$$

$$\rightarrow (k^2 - 15k)(-4k + 60)$$

$$k(k - 15) - 4(k - 15)$$

25. $4m^2 - m - 14$

$$m^2 - m - 56$$

$$(m - 2)(4m + 7)$$

26. $8r^2 + 16r - 10$

$$2(4r^2 + 8r - 5)$$

$$(4r^2)$$

$$2(2r + 5)(2r - 1)$$

27. If the area of a rectangle can be represented by $(x^2 - 5x - 14)$, what two expressions could represent the dimensions of the rectangle?

$$(x^2 - 7x)(x + 2)$$

$$x(x - 7) + 2(x - 7)$$

$$(x - 7)(x + 2)$$

Topic 6: Dividing a Polynomial by a Binomial

Directions: Find each quotient.

28. $(x^2 - 14x + 48) \div (x - 6)$

$$(x - 6)(x - 8)$$

$$(x - 6)$$

$$x - 8$$

29. $(p^2 - 9) \div (p - 3)$

$$(p + 3)(p - 3)$$

$$(p - 3)$$

$$p + 3$$

30. $\frac{a^2b + a^2c - 5b - 5c}{b + c}$

$$a^2(b + c) - 5(b + c)$$

$$\frac{(a^2 - 5)(b + c)}{(b + c)} = a^2 - 5$$